

DS Practical - 9

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PART B - QUESTIONS

Searching Algorithms:

Write programs to implement and compare:

a. Linear search

CODE :

```
def linearSearch(array, n, x):
    for i in range(0, n):
        if (array[i] == x):
            return i
    return -1

array = [2, 4, 0, 1, 9]
x = 1
n = len(array)
result = linearSearch(array, n, x)
if(result == -1):
    print("Element not found")
else:
    print("Element found at index: ", result)
```

OUTPUT :

```
----- REGIAR. C./0815/1
Element found at index: 3
```

b. Binary search (on a sorted array)

CODE :

Iterative Method

```

def binarySearch(array, x, low, high):
    while low <= high:
        mid = low + (high - low)//2

        if x == array[mid]:
            return mid
        elif x > array[mid]:
            low = mid + 1
        else:
            high = mid - 1
    return -1

array = [3, 4, 5, 6, 7, 8, 9]
x = 4
result = binarySearch(array, x, 0, len(array)-1)

if result != -1:
    print("Element is present at index " + str(result))
else:
    print("Not found")

```

OUTPUT :

```

Element is present at index 1
|

```